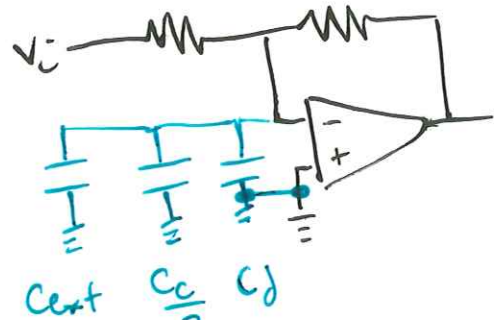
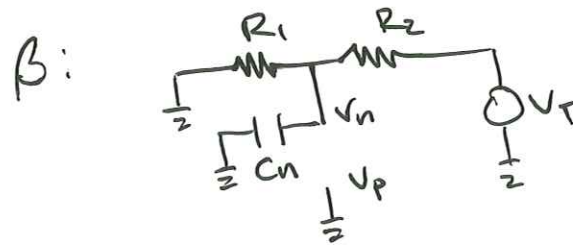


Continued Stability



all in $\parallel$, $\rightarrow C_n = C_{ext} \parallel \frac{C_c}{2} \parallel C_d$
 $C_n = C_{ext} + \frac{C_c}{2} + C_d$

$A_{ideal} \rightarrow V_n \rightarrow 0$ (virtual short)
 $\therefore C_n$ has no impact on A_{ideal}
 $A_{ideal} = -R_2/R_1$



$R_1 \rightarrow R_1 \parallel Z_{Cn}$

$\beta_{f=0} = \frac{R_1}{R_1 + R_2}$ $\beta_{f \neq 0} = \frac{R_1 \parallel Z_{Cn}}{R_1 \parallel Z_{Cn} + R_2}$

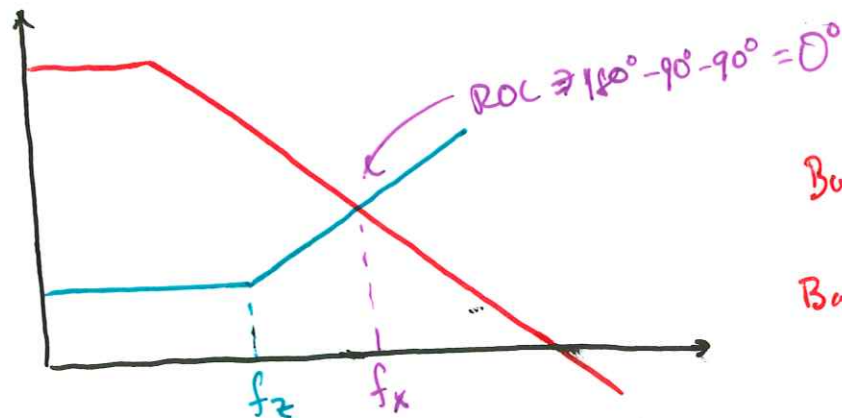
for ROC analysis, want

$1/\beta(jf) = \left(1 + \frac{R_2}{R_1}\right) \left(1 + jf/f_z\right)$

$f_z = 1/2\pi(R_1 \parallel R_2)C_n$

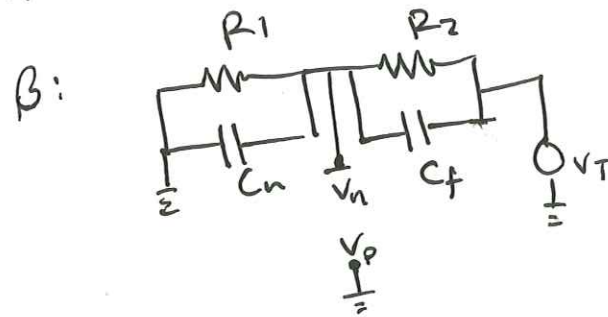
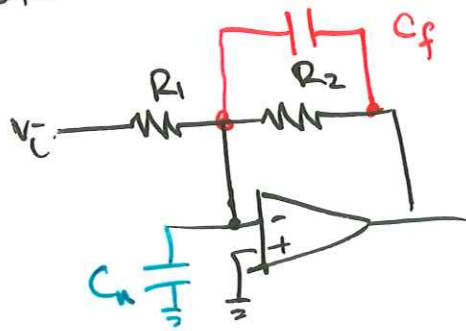
$f \rightarrow 0 \quad 1/\beta \rightarrow 1 + \frac{R_2}{R_1}$

$f \rightarrow \infty \quad |1/\beta| \rightarrow \infty$



Bad solution ①: Add more C_{ext}
 more f_z very low
 Bad solution ②: Manipulate $R_1 \parallel R_2$
 to move f_z very low
 or very high.

Better solution: Add a pole to $1/\beta$



$$\beta = \frac{R_1 \parallel Z_{Cn}}{R_1 \parallel Z_{Cn} + R_2 \parallel C_f}$$

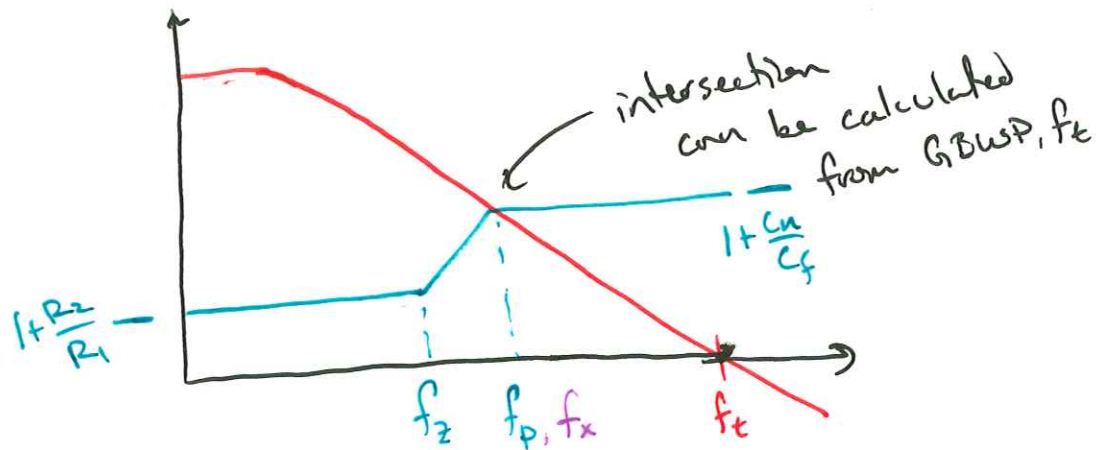
$$1/\beta|_{f=0} = 1 + \frac{R_2}{R_1}$$

$$1/\beta|_{f \rightarrow \infty} = 1 + \frac{Z_{Cf}}{Z_{Cn}} = 1 + \frac{C_n}{C_f}$$

$$1/\beta = 1 + \frac{R_2 \parallel Z_{Cf}}{R_1 \parallel Z_{Cn}} = \left(1 + \frac{R_2}{R_1}\right) \frac{1 + jf/f_z}{1 + jf/f_p}$$

$$f_z = 1/2\pi(R_1 \parallel R_2)(C_n + C_f)$$

$$f_p = 1/2\pi R_2 C_f$$



$$f_t = f_b / |a_{ol}|$$

$$f_t = f_p |1/\beta(f_p)|$$

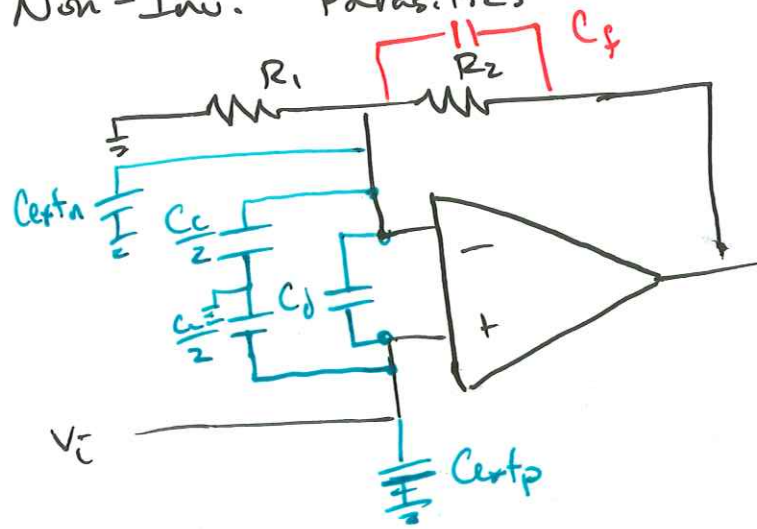
$$f_t \approx f_p \left(1 + \frac{C_n}{C_f}\right)$$

$$C_f = \frac{(1 + \sqrt{1 + 8\pi R_2 C_n f_t})}{4\pi R_2 f_t}$$

f_p could also be much to f_z such that intersection results in $\text{POC} \rightarrow \phi_m \approx 90^\circ$ significant drop in f_B !

(ex $f_z = f_p$, pole-zero cancellation.)

Non-Inv. Parasitics



if V_i source is low-impedance or ideal, then C_{extn} , $\frac{C_c}{2}$ (on + terminal) may be ignored!

$$1/\beta(j\omega) = \left(1 + \frac{R_2}{R_1}\right) \frac{1 + j\omega/\omega_z}{1 + j\omega/\omega_p}$$

$$\omega_z = 1/2\pi(R_1 \parallel R_2)(C_f + C_n)$$

$$\omega_p = 1/2\pi R_2 C_f$$

$$C_n = C_{extn} + \frac{C_c}{2} + C_d$$

Same from previous!

$$A_{ideal} \quad f \rightarrow 0 \quad A_{ideal} = 1 + \frac{R_2}{R_1}$$

$$f \neq 0 \quad A_{ideal}|_{f \neq 0} = \left(1 + \frac{R_2}{R_1}\right) \left[\frac{1 + j2\pi f(R_1 \parallel R_2)(C_f + C_x)}{1 + j2\pi f R_2 C_f} \right]$$

$$C_x = C_{extn} + \frac{C_c}{2}$$

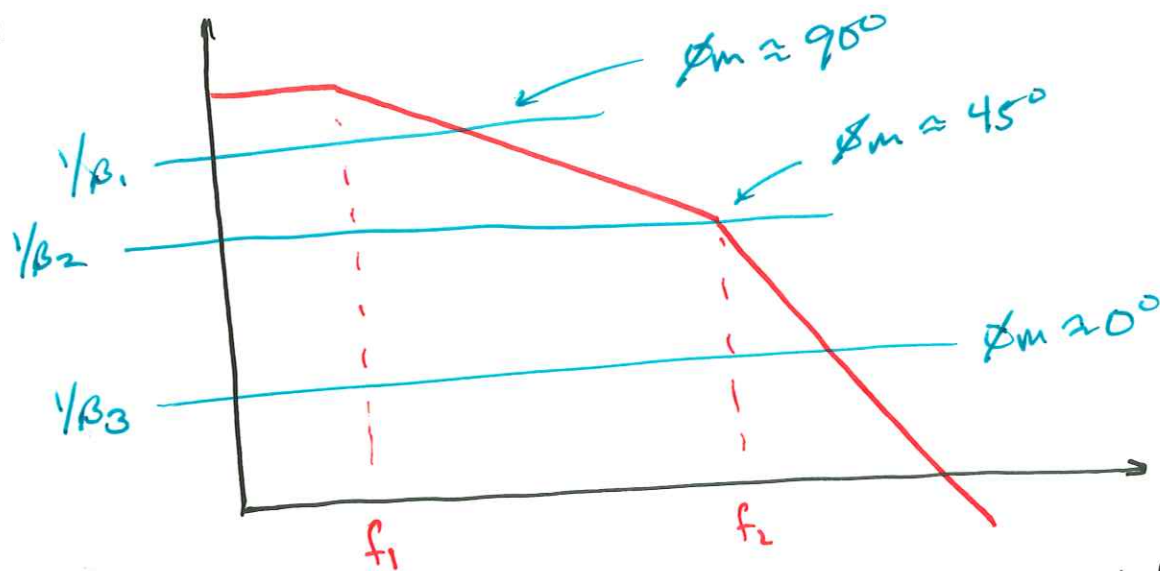
$$\text{want } A_{ideal}|_{f \neq 0} = \left(1 + \frac{R_2}{R_1}\right)$$

$$\text{need: } R_1 \parallel R_2 (C_f + C_x) = R_2 C_f$$

$$\rightarrow C_f = \frac{R_1}{R_2} C_x$$

Non-Dominant Pole amps (Non-GBWP)

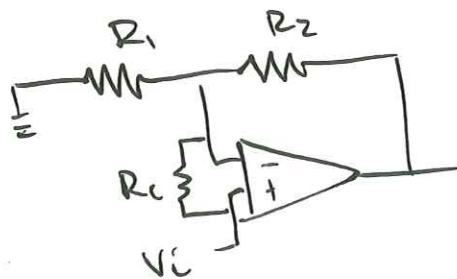
ex



Solution ①: increase $1/\beta$ to provide stability.
implies [very] high gain

Assume parasitics are negligible...

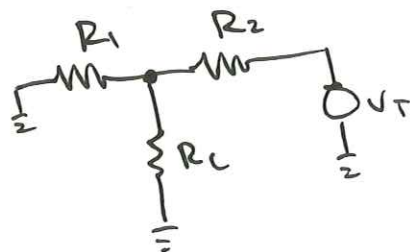
Solution (2)



Since R_c across V_n, V_p
does not affect A_{ideal}

$$A_{ideal} = 1 + \frac{R_2}{R_1}$$

$1/\beta$



$$\beta = \frac{R_1 \parallel R_c}{R_1 \parallel R_c + R_2}$$

$$1/\beta = 1 + \frac{R_2}{R_1} + \frac{R_2}{R_c}$$

want $\phi_m = 45^\circ$

$$1/\beta = |a(jf_2)| \rightarrow R_c = \frac{R_2}{(|a(jf_2)| - (1 + \frac{R_2}{R_1}))}$$

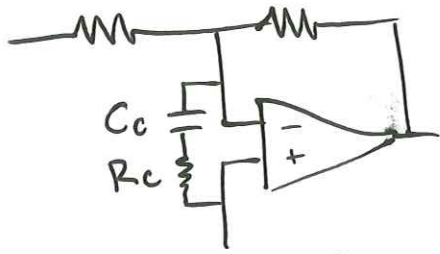
since we increase $1/\beta$ and $T = a\beta$

then T significantly reduced!

→ significantly increases gain error!

→ lower bandwidth.

Solution ③: Improve upon ②



Aideal still unchanged!

R_c from previous now becomes:

$$R_c + Z_{Cc} = R_c + \frac{1}{j2\pi f C_c}$$

$$\frac{1}{B(jf)} = \left(1 + \frac{R_2}{R_1}\right) \frac{1 + jf/f_z}{1 + jf/f_p}$$

$$f_z = \frac{1}{2\pi(R_c + R_1 \parallel R_2) C_c}$$

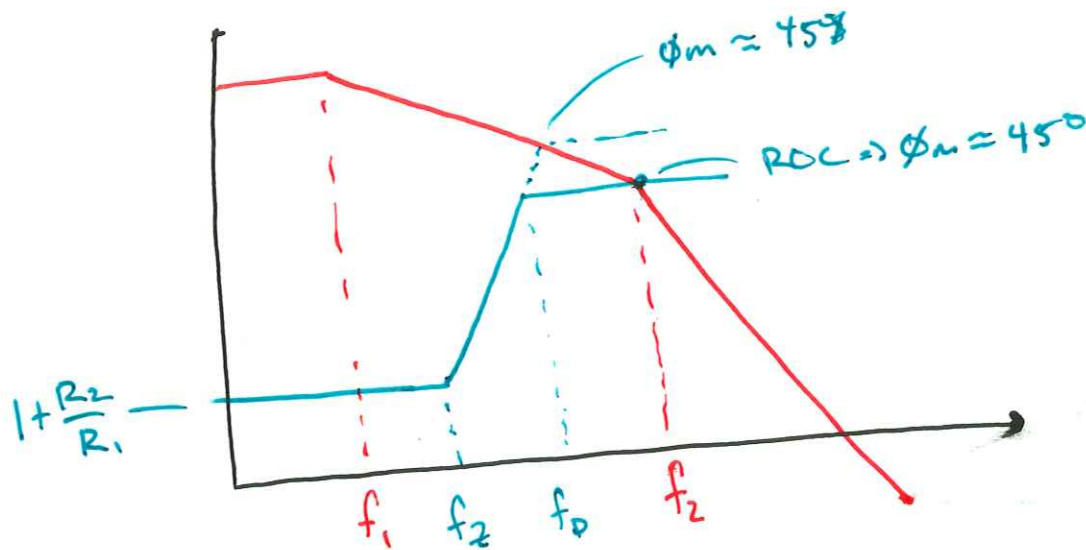
$$f_p = \frac{1}{2\pi R_c C_c}$$

$$f_z < f_p$$

$$(R_c + R_1 \parallel R_2) > R_c$$

$$\frac{1}{B}|_{f \rightarrow 0} = 1 + \frac{R_2}{R_1}$$

$$\frac{1}{B(jf)}|_{f \rightarrow \infty} = 1 + \frac{R_2}{R_1} + \frac{R_2}{R_c}$$



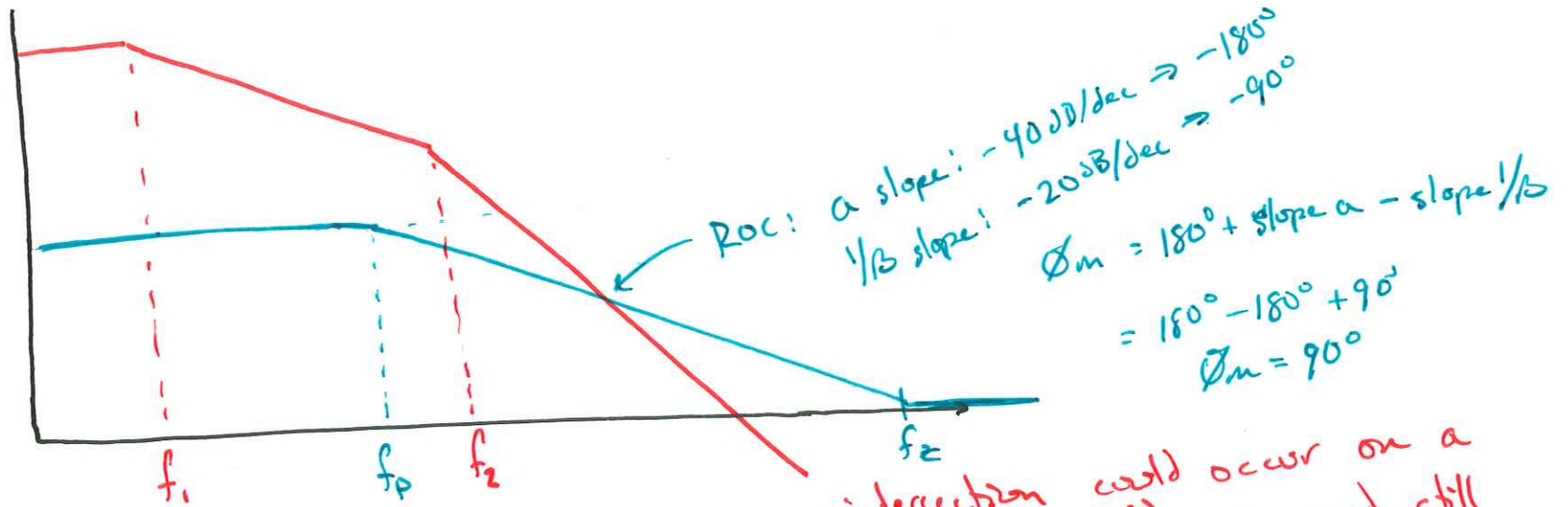
$$\text{set } f_p = 0.1 f_z$$

$$\rightarrow C_c = 5/\pi R_c f_z$$

Solve for R_c from previous solution.

Solution (4)

Add pole to $1/\beta$ to decrease ROC at high f .
 $\rightarrow$ Add feedback Cap, C_f



intersection could occur on a 3rd pole (f_3) in a and still be stable!

$$1/\beta = \left(1 + \frac{R_2}{R_1}\right) \frac{1 + jf/f_z}{1 + jf/f_p}$$

$$f_z = \frac{1}{2\pi(R_1 \parallel R_2)C_f}$$

$$f_p = \frac{1}{2\pi R_2 C_f} \quad (1)$$

$$f_p < f_z$$

Step 1: $-f_n$ from (3)

- get f_p from (2)

- get C_f from (1)

if no f_3 , then select f_p such that $f_p < 0.1 f_x$

if f_3 exists, want f_3 exactly in between $f_z \leq f_p$

$$f_x = f_3 = \sqrt{f_p f_z} = f_p \sqrt{1 + \frac{R_2}{R_1}} \quad (2)$$

$$|a(jf_x)| = |1/\beta(jf_x)| = \sqrt{1 + \frac{R_2}{R_1}} \quad (3)$$

$$\phi_m = 180^\circ + \angle a(jf_x) - 90^\circ + 2 \tan^{-1} \sqrt{1 + \frac{R_2}{R_1}}$$